Anirudha N Shastri

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RESEARCH PROFILE

Medical AI researcher with experience translating clinical questions into reproducible computational studies across interventional oncology, medical imaging, and optical spectroscopy. Build end-to-end research systems spanning clinical data curation, DICOM processing, model development, 3D image analysis, quantitative evaluation, and scientific communication. Research interests include grounded and interpretable medical AI, image-guided procedures, multimodal clinical data, and clinically meaningful model evaluation.

EDUCATION

Northeastern University, Boston, MA — August 2022 - December 2024
M.S. in Artificial Intelligence, Khoury College of Computer Sciences

Dayananda Sagar University, Bengaluru, India — August 2018 - June 2022
B.Tech. in Computer Science and Engineering

RESEARCH EXPERIENCE

Memorial Sloan Kettering Cancer Center, New York, NY — January 2025 - Present
Senior Research Technician, Department of Interventional Radiology

Conduct translational AI research with interventional radiology teams across image registration, tumor segmentation, lesion detection, and biopsy spectroscopy; contribute from clinical data preparation and experimental design through quantitative evaluation and scientific dissemination.

Foundation Deformable Registration - Lung Thermal Ablation

- Built a lung CT registration workflow around foundational uniGradICON, systematically testing preprocessing and masking strategies - including whole-lung and localized ROI masks and fine-tuning- to determine configurations that improved robustness on institutional pre/post-ablation data.
- Benchmarked B-spline against the foundational uniGradICON model using radiologist-defined vessel-bifurcation target registration error (TRE), instance optimization, and dataset-specific tuning; best B-spline TRE was 0.43 mm versus 1.41 mm for uniGradICON, with an approximately 10x speed tradeoff.
- Curated the retrospective research cohort from raw institutional imaging, building reproducible anonymization and DICOM/DICOM-SEG-to-NRRD workflows and resolving correspondence and geometry issues required for valid longitudinal registration and downstream 3D margin computation.

Human-Assisted Diffusion Models - Lung Tumor Segmentation

- Developed a human-assisted diffusion segmentation workflow in which a user-selected tumor center defines a focused ROI, investigating whether lightweight interaction can make diffusion-based segmentation practical for time-sensitive clinical workflows.
- Achieved DDPM Dice scores of 0.73 on the Medical Segmentation Decathlon dataset and 0.92 on an institutional 2D dataset; extended the framework to 3D with DDIM, achieving 0.77 Dice using only 20 denoising steps rather than the 1,000-step DDPM formulation.
- Demonstrated that ROI-based inference reduced mean 3D runtime from 5.28s to 0.90s at 20 DDIM steps while preserving segmentation performance; also evaluated cross-dataset generalization and diffusion-derived uncertainty maps around tumor boundaries.

Optical Spectroscopy - Biopsy Tissue Characterization

- Recommissioned a custom in-house optical spectroscopy platform - the sole system of its design - developed roughly a decade earlier; reconstructed its operating workflow from legacy hardware and LabVIEW code after formal documentation was no longer available, independently learning the optics and LabVIEW needed to troubleshoot and calibrate the system.
- Co-designed a 3D-printed biopsy-sample holder by translating physician workflow needs and spectral-acquisition geometry into design requirements; collaborated with medical physics through iterative prototyping to improve usability and acquisition consistency.
- Supported ex vivo biopsy spectroscopy and calibration experiments; developing signal-processing and machine-learning approaches for tissue characterization and exploring whether rapid spectral assessment could inform specimen adequacy for downstream molecular analyses such as sequencing.

Fracture & Lesion Detection - Pelvic Radiographs

- Built an end-to-end Detectron2 Faster R-CNN pipeline for radiologist-annotated pelvic radiographs, spanning preprocessing, training, controlled inclusion of normal images, held-out testing, and COCO-style performance/error analysis.
- Found that adding 20% normal radiographs improved held-out overall mAP from 13.3 to 14.5 and AP75 from 2.0 to 11.6; fracture detection reached AP50 of 45.0, outperforming lesion detection.

PUBLICATIONS & PATENTS

- **RSNA Radiology: Cancer Imaging 2026 —** Automated Deformable Registration and Three-dimensional Margin Assessment for Predicting Local Recurrence after Lung Thermal Ablation
- **SPIE Medical Imaging 2026 —** Interactive Lung Tumor Segmentation using Denoising Diffusion Probabilistic Models
- **SPIE Medical Imaging 2026 —** Comparing B-spline and uniGradICON AI Foundational Deformable Image Registration Methods: Evaluating the Tradeoff Between Accuracy and Speed
- **RSNA 2025 —** AI-based automated detection and characterization of bone lesions and fractures on pelvic radiographs
- **Indian Patent Office, 2025 —** Automated Cervical Vertebral Staging to Determine Skeletal Maturation Using Lateral Cephalograms. Application No. 202541002362.

WORK EXPERIENCE

Unravel Bioscience, Boston, MA — June 2023 - December 2023
Computational Drug Discovery Intern

- Re-engineered a computer-vision tadpole tracking system from group-level to individual-level tracking, enabling longitudinal behavioral analysis across experiments.
- Expanded validated behavioral metrics from 3 to 7, modularized computer-vision and post-processing components for scale, and built an ECG-like diagnostic visualization for movement phenotyping.

nference, Bengaluru, India — February 2022 - April 2022
Computer Vision Intern

- Designed an OCR pipeline using CRAFT and Tesseract to digitize handwritten labels from legacy pathology slides, supporting large-scale pathology data curation.
- Developed a computer-vision workflow for KI-67 marker detection in breast cancer pathology to reduce manual nuclei-counting effort.

ACADEMIC RESEARCH

AI Dental Diagnostics & Cervical Vertebral Staging | Dayananda Sagar University / Northeastern — 2021 - 2024

- Led a 5-person team developing an AI-assisted dental diagnostics platform; collaborated with dental clinicians to define research requirements, curated more than 2,500 radiographs, and built a custom annotation tool for 400 labeled images.
- Developed landmark-localization and staging models using hourglass networks, U-Net, and InceptionResNetV2; the later U-Net system reached 85% PCK with 0.25 Dice loss and improved localization over the earlier framework.
- Project won 1st place among 97 projects at the DSU Project Expo 2022, contributed to a university Dental College collaboration/MOU, and led to a published Indian patent application.

TECHNICAL & RESEARCH SKILLS

Programming: Python, C/C++, MATLAB, SQL, Java, JavaScript

Machine Learning / Computer Vision: PyTorch, TensorFlow, MONAI, Detectron2, OpenCV; segmentation, deformable registration, object detection, landmark localization, transfer learning, hyperparameter optimization

Medical Imaging / Clinical Data: SimpleITK, DICOM, DICOM-SEG, NRRD, 3D image geometry, image resampling, anonymization, dataset curation, clinical imaging QA

Research / Analysis: experimental design, model evaluation, Dice, TRE, COCO AP/mAP, PR/ROC analysis, spectroscopy signal processing, scientific visualization, literature-driven method development

Tools / Platforms: Git, Linux, GCP, AWS, 3D Slicer